Parth N. Kheni

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EDUCATION

Boston University — **Boston, MA**

B.S. Computer Engineering, Concentration in Machine Learning — *Expected Dec 2026*

- Related Coursework: *Machine Learning, Probability Statistics & Data Science, Computational Linear Algebra, Applied Algorithms, Cloud Computing, Software Engineering, Computer Architecture, Differential Equations*

Rutgers University — **New Brunswick, NJ**

B.S. Computer Engineering — *(Transferred) Sept 2023 – May 2024*

SKILLS

Programming: Python, SQL, MATLAB, C++ (primary), Java, C#, Objective-C, OOP
Libraries/Frameworks: Numpy, Matplotlib, Pandas, scikit-learn, TensorFlow, PyTorch
Cloud/Tools: Azure AD, GitHub, Microsoft 365, Arduino, Android Studio, REST APIs, Power BI, WorkManager
Systems & Analysis: Systems Integration, Cross-Team Coordination, Time-Series Analysis, Predictive Modeling

EXPERIENCE

Tufts University AABL Lab (CRA-WP DREU), *Undergraduate Research Fellow* — *June 2026 – Present*

- Selected for the competitive, nationally funded CRA-WP Distributed Research Experiences for Undergraduates fellowship to conduct research with Prof. Elaine Short in socially assistive human-robot interaction and human-in-the-loop machine learning.

New Harbor Venture Partners, *Applied Data Science Intern* — *May 2026 – Present*

- Migrating a ~20,000-contact investor CRM (Affinity) by building a Python pipeline to clean, deduplicate (email-keyed), and import PitchBook exports into standardized fund/contact records for targeted outreach.
- Designing an LLM workflow that connects Claude to the CRM via a Model Context Protocol (MCP) to auto-tag funds by investment criteria (stage, check size, sector) and surface ranked investor targets per deal.

Boston University Center for Space Physics, *Machine Learning Researcher – Space Weather* — *Apr 2025 – June 2026*

- Built end-to-end Python pipelines for POES satellite instrument data and OMNI solar-wind measurements, generating model-ready datasets of 7M+ 2-second samples/month for space-weather forecasting.
- Trained and evaluated PyTorch neural networks and Ridge regressors to forecast energetic electron precipitation (E4_0/E4_90), improving RMSE and correlation over prior baselines with a focus on generalization across storm conditions and rare high-flux events.

Beth Israel Deaconess Medical Center (BIDMC), *Machine Learning Intern* — *Sept 2025 – May 2026*

- Led a multi-institutional collaboration between BIDMC and Johns Hopkins across engineering and clinical teams to develop and integrate a learning-based autonomous wound closure algorithm within the da Vinci Research Kit (dVRK) platform for future clinical translation, coordinating requirements and experiment milestones.
- Trained Action Chunking Transformer + ResNet-18 gesture-classification models on 1,890 expert demonstrations (10 tissue phantoms), achieving 0.707 mm positional error below the 1.0 mm clinical threshold and <0.99 mm RMSE on unseen tissue.
- Annotated 853 expert trajectories with 16 surgical gesture-phase labels and ran systematic ablations and out-of-distribution tissue experiments, isolating training-data diversity as the key generalization bottleneck.

PUBLICATIONS & PRESENTATIONS

- Capannolo, L., Pettit, J., Elliott, S., Morales, J. P. B., Bibeau, A., **Kheni, P. N.** — *Feb 2026*
  "Monitoring radiation belt electrons at LEO: preliminary results of MAPLE (MAPs of LEO Electrons) model."
  COSPAR 2026 – Abstract (submitted)

PROJECTS

LLM-Assisted Incident Summarization & Clustering (Microsoft Research), *Applied ML Engineer* — *Jan 2026 – Present*

- Build an end-to-end incident intelligence pipeline: ingest large-scale system/application logs, generate structured incident summaries with LLMs, embed summaries, and cluster incidents to identify duplicates and failure patterns.
- Implement a backend service and lightweight UI/API to explore incidents and clusters, making practical accuracy vs cost/latency tradeoffs to support scalable operation on noisy real-world logs.

Whack-a-Mole Reaction Game on FPGA (Nexys4 DDR, Verilog), *FPGA Design Engineer* — *Sept 2025 – Dec 2025*

- Built a real-time Whack-a-Mole game on a Nexys4 DDR FPGA in Verilog (debounced I/O, timing, randomized 30s rounds), owning the input/game-control modules and delivering a glitch-free demo verified through Vivado simulation.

AWARDS & HONORS

NSF ACCESS / Jetstream2 Grant (CIS260066) – $31,444 awarded; 146,000 GPU SUs. — *Jan 2026*

GEM Workshop – Invited Speaker (Portland, ME) – selected to present machine learning research. — *July 2026*

BU Undergraduate Research Opportunities Program (UROP) Award – space weather ML research. — *Summer 2026*

ACTIVITIES

BU Baja Society of Automotive Engineers, *Head Engineer of Computer Systems* — *Sept 2024 – Present*

BU Jalwa Dance Troupe, *Secretary* — *Sept 2024 – Present*